

A Lifted Primal-Domain Formula for the Petz–Augustin Random-Coding Exponent

Run `fa_banach_001` — candidate solution packet

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Status. Candidate full solution, likely valid, scoped to finite input alphabets and finite-dimensional output systems. The formula is an exact spectral lift with finite time-sharing. A quantum-state-only primal formula is not claimed.

1 Source question

Cheng and Liu prove that the constant-composition random-coding exponent of a classical–quantum channel $\mathcal{N} : x \mapsto \rho_x$ with fixed input law p is

$$E_{\text{P}}(R) = \sup_{\alpha \in [1/2, 1]} \frac{1 - \alpha}{\alpha} [I_{\alpha}^{\text{Aug}}(p; \mathcal{N}) - R], \quad (1)$$

where I_{α}^{Aug} is built from the Petz Rényi divergence [1, Theorem 4.3]. In Remark 4.4, PDF page 23, they explain why the literal quantum analogue of the classical Csiszár–Körner formula yields the wrong, log-Euclidean Rényi divergence, and ask whether any primal-domain expression exists. The complete source context is reproduced below.

Theorem 4.3 (Random coding bound for constant-composition codes). *Let $\mathcal{N}_{\mathbf{X} \rightarrow \mathbf{B}} : x \mapsto \rho_{\mathbf{B}}^x$ be a classical-quantum channel. For any integer $n \in \mathbb{N}$ for which $p_{\mathbf{X}}$ is an n -type, and any rate $R := \frac{1}{n} \log_2 |\mathbf{M}|$, the following bound holds for the constant-composition code $\tilde{p}_{\mathbf{X}^n}$ in (76):*

$$\log_2 \varepsilon(\mathbf{X}^n : \mathbf{B}^n)_{\tilde{p}} \leq -n \frac{1-\alpha}{\alpha} [I_{\alpha}^{\text{Aug}}(p; \mathcal{N}) - R] + \frac{|\mathbf{X}|}{\alpha} \log_2(n+1) + \log_2 c_{\alpha}, \quad \forall \alpha \in [1/2, 1]. \quad (78)$$

Here, the order- α Petz–Augustin information $I_{\alpha}^{\text{Aug}}(p; \mathcal{N})$ is defined in (73) with respect to the state $\rho_{\mathbf{XB}} = \sum_{x \in \mathbf{X}} p_{\mathbf{X}}(x) |x\rangle\langle x|_{\mathbf{X}} \otimes \rho_{\mathbf{B}}^x$.

Moreover, for $R \geq \frac{d}{ds} s I_{\frac{1}{1+s}}^{\text{Aug}}(p; \mathcal{N})|_{s=1}$,

$$-\log_2 \varepsilon(\mathbf{X}^n : \mathbf{B}^n)_{\tilde{p}} = n \cdot \sup_{\alpha \in [1/2, 1]} \frac{1-\alpha}{\alpha} [I_{\alpha}^{\text{Aug}}(p; \mathcal{N}) - R] + \mathcal{O}(\log_2 n) \quad (79)$$

(where the exact factors in $\mathcal{O}(\log_2 n)$ can be found in (78) and in the converse bound [22]).

Remark 4.4. In the commuting case, Theorem 4.3 recovers (the non-asymptotic version of) Csiszár–Körner’s random coding bound [55, Theorem 10.2]. Note that our result in terms of the parametric Rényi divergence, $\sup_{\alpha \in [1/2, 1]} \frac{1-\alpha}{\alpha} [I_{\alpha}^{\text{Aug}}(p; \mathcal{N}) - R]$, is commonly called the *dual-domain* expression [130], while Csiszár–Körner’s result is expressed in the *primal domain* (which matches the Haroutunian form of the sphere-packing bound [131]):

$$\inf_{\mathcal{M}_{\mathbf{X} \rightarrow \mathbf{B}} : x \mapsto \sigma_{\mathbf{B}}^x} \left\{ \mathbb{E}_{x \sim p_{\mathbf{X}}} D_1(\sigma_{\mathbf{B}}^x \| \rho_{\mathbf{B}}^x) + \left(I_1^{\text{Aug}}(p; \mathcal{M}) - R \right)_+ \right\}. \quad (80)$$

However, in the noncommuting case, the dual-domain expression of (80) corresponds to the Augustin information defined via the *log-Euclidean* Rényi divergence [22, 53, 132], instead of the Petz–Rényi divergence. This means that the random coding error exponent of the form (80) is generally not achievable because of the sphere-packing bound established in [21, 22]. In light of this, our result in Theorem 4.3 is the only such expression currently known to us; we do not know if there is a primal-domain expression as in the classical case.

2 Definitions

All logarithms and relative entropies below use base two. Let $\mathcal{X}$ be finite, let $\mathcal{H}$ be finite-dimensional, let p be a probability distribution on $\mathcal{X}$, and let $\rho_x \in \mathcal{S}(\mathcal{H})$. For $0 < \alpha < 1$,

$$D_{\alpha}(\rho \| \sigma) = \frac{1}{\alpha - 1} \log \text{Tr}(\rho^{\alpha} \sigma^{1-\alpha}), \quad I_{\alpha}^{\text{Aug}}(p; \mathcal{N}) = \inf_{\sigma \in \mathcal{S}(\mathcal{H})} \sum_x p(x) D_{\alpha}(\rho_x \| \sigma).$$

Fix spectral decompositions

$$\rho_x = \sum_i r_{x,i} |u_{x,i}\rangle\langle u_{x,i}|, \quad \sigma = \sum_j q_j |v_j\rangle\langle v_j|.$$

The associated Nussbaum–Szkola distributions [2, 3] on the pairs (i, j) are

$$P_{x,\sigma}(i, j) = r_{x,i} |\langle u_{x,i}, v_j \rangle|^2, \quad Q_{x,\sigma}(i, j) = q_j |\langle u_{x,i}, v_j \rangle|^2. \quad (2)$$

A *primitive primal datum* is $\omega = (\sigma, (T_x)_{x \in \mathcal{X}})$, where σ is faithful and each T_x is a probability distribution on the eigen-index pairs. Define extended nonnegative costs

$$A(\omega) = \sum_x p(x) D(T_x \| P_{x,\sigma}), \quad B(\omega) = \sum_x p(x) D(T_x \| Q_{x,\sigma}). \quad (3)$$

Only data with finite costs are needed. Let

$$\mathcal{K} = \text{cl conv} \{ (A(\omega), B(\omega)) : \omega \text{ is a finite-cost primitive datum} \} \subseteq \mathbb{R}^2. \quad (4)$$

Thus $\mathcal{K}$ permits ordinary finite time-sharing and limiting sequences, but it does not depend on the Rényi order.

3 Proof Intuition

The obstacle identified by the source is noncommutativity: the usual Gibbs variational argument applied directly to matrices creates the log-Euclidean Rényi divergence. The key idea is to apply Gibbs variation only after the Nussbaum–Szkoła spectral lift. That lift preserves the Petz Rényi overlap exactly, while turning it into an ordinary classical overlap.

For $s = (1 - \alpha)/\alpha$, the classical tilted distribution converts each term sD_α into a minimum of $A + sB$. Minimizing over the Augustin state therefore identifies the dual exponent with $\sup_{0 \leq s \leq 1} \inf_{(A,B) \in \mathcal{K}} [A + s(B - R)]$. Convex time-sharing makes Sion minimax applicable. After exchanging the two optimizations, maximizing over s gives precisely the hinge $(B - R)_+$. In the commuting case, A is conditional relative entropy and B reduces to mutual information, recovering the classical formula.

4 Main result

Theorem 1 (Lifted primal formula). *For every finite classical–quantum channel $\mathcal{N} : x \mapsto \rho_x$, input law p , and rate $R \in \mathbb{R}$, the Petz–Augustin exponent in (1) satisfies*

$$E_P(R) = \inf_{(A,B) \in \mathcal{K}} \{A + (B - R)_+\}. \quad (5)$$

Every point relevant to this infimum can be approximated by time-sharing among at most three primitive data. If all ρ_x commute, (5) reduces to the Csiszár–Körner formula

$$\inf_{V: x \mapsto V_x} \left\{ \sum_x p(x) D(V_x \| W_x) + (I(p, V) - R)_+ \right\}, \quad (6)$$

where W_x are the common-basis eigenvalue distributions of ρ_x .

5 Proof

Lemma 1 (Classical tilt identity). *For probability distributions P, Q on a finite set and $s \geq 0$, put $\alpha = 1/(1 + s)$. Then*

$$sD_\alpha(P \| Q) = \inf_T \{D(T \| P) + sD(T \| Q)\}. \quad (7)$$

For $s > 0$ the minimizer is

$$T_s(z) = \frac{P(z)^\alpha Q(z)^{1-\alpha}}{\sum_y P(y)^\alpha Q(y)^{1-\alpha}}.$$

The equality at $s = 0$ is understood by continuity and is also immediate.

Proof. Let $Z_s = \sum_z P(z)^{1/(1+s)} Q(z)^{s/(1+s)}$. For any admissible T ,

$$D(T \| P) + sD(T \| Q) = (1 + s)D(T \| T_s) - (1 + s) \log Z_s.$$

The first term has minimum zero. The second equals $sD_{1/(1+s)}(P \| Q)$ by definition. $\square$

Lemma 2 (Petz preservation). *The distributions in (2) satisfy*

$$D_\alpha(P_{x,\sigma} \| Q_{x,\sigma}) = D_\alpha(\rho_x \| \sigma), \quad 0 < \alpha < 1.$$

Proof. Writing $c_{ij} = |\langle u_{x,i}, v_j \rangle|^2$, one has

$$\begin{aligned} \sum_{i,j} P_{x,\sigma}(i,j)^\alpha Q_{x,\sigma}(i,j)^{1-\alpha} &= \sum_{i,j} r_{x,i}^\alpha q_j^{1-\alpha} c_{ij} \\ &= \text{Tr}(\rho_x^\alpha \sigma^{1-\alpha}). \end{aligned}$$

Taking $(\alpha - 1)^{-1} \log$ proves the claim. $\square$

Proof of Theorem 1. Write $s = (1 - \alpha)/\alpha$, so $\alpha \in [1/2, 1]$ corresponds to $s \in [0, 1]$. For $s > 0$, Lemmas 1 and 2, followed by separate minimization over each T_x , give

$$\begin{aligned} sI_{1/(1+s)}^{\text{Aug}}(p; \mathcal{N}) &= \inf_{\sigma > 0} \sum_x p(x) s D_{1/(1+s)}(\rho_x \| \sigma) \\ &= \inf_{\omega} \{A(\omega) + sB(\omega)\} \\ &= \inf_{(A,B) \in \mathcal{K}} \{A + sB\}. \end{aligned} \tag{8}$$

Restricting the first infimum to faithful σ does not change its value: replace any σ by $(1 - \varepsilon)\sigma + \varepsilon I / \dim \mathcal{H}$ and let $\varepsilon \downarrow 0$. The required Petz overlaps are continuous for $\alpha < 1$. At $s = 0$, both sides of (8) equal zero.

Substitute (8) into (1):

$$E_P(R) = \sup_{s \in [0,1]} \inf_{(A,B) \in \mathcal{K}} \{A + s(B - R)\}.$$

The set $[0, 1]$ is compact convex, $\mathcal{K}$ is convex, and the displayed function is continuous affine in both variables. Apply Sion's minimax theorem to its negative (with s as the compact minimization variable) to obtain

$$\begin{aligned} E_P(R) &= \inf_{(A,B) \in \mathcal{K}} \sup_{s \in [0,1]} \{A + s(B - R)\} \\ &= \inf_{(A,B) \in \mathcal{K}} \{A + (B - R)_+\}, \end{aligned}$$

which is (5). Since $\mathcal{K} \subseteq \mathbb{R}^2$, Carathéodory's theorem says that every point of its nonclosed convex hull is a convex combination of at most three primitive cost pairs; closure points are limits of such combinations.

It remains to verify the commuting reduction. In a common eigenbasis, each spectral lift is an ordinary classical refinement of the channel W_x . A primitive T_x can be pushed through the map that forgets the refinement, giving a channel V_x . Data processing yields

$$\sum_x p(x) D(V_x \| W_x) \leq A, \quad I(p, V) \leq B.$$

For the unrefined diagonal choice, the sharper chain rule is

$$\sum_x p(x) D(V_x \| q) = I(p, V) + D(pV \| q),$$

so minimizing over q gives $B = I(p, V)$. For a time-sharing mixture, average the resulting channels V ; joint convexity of relative entropy and convexity of $I(p, V)$ show that this averaging does not increase either relevant cost. Thus no lifted/time-shared datum beats (6), while every candidate in (6) occurs as a single diagonal primitive datum. The two infima are equal. $\square$

6 Verification, scope, and novelty

The script `code/check_variational_identity.py` performs two checks:

- 400 seeded random noncommuting matrix/order instances of Lemma 1 combined with Lemma 2;
- 100 seeded random instances of the classical chain rule used in the commuting reduction.

The command is

```
conda run --no-capture-output -n sandbox python \
  code/check_variational_identity.py
```

The computation is a sanity check, not part of the proof.

The result is exact in the finite setting surrounding source Remark 4.4. It does not supply an infinite-dimensional extension. More importantly, its primal variables are classical tilted laws on canonical spectral lifts, with finite

time-sharing. Whether the same exponent admits a comparably simple formula involving only alternative quantum output states remains open.

The bounded novelty search covered the four run indexes; exact-phrases arXiv searches for “primal-domain expression” with Petz–Augustin information; Nussbaum–Szkola, Csiszár–Körner, sphere-packing, and constant-composition variants; arXiv:2308.02929; and arXiv:2601.06492. No matching exponent formula or answer to Remark 4.4 was found. Novelty confidence is therefore provisional rather than definitive.

Human-review recommendation. Review as a scoped candidate full solution. In particular, verify the faithful-state approximation, the sign in the Sion step, and whether a spectral-lift formula meets the intended breadth of the source’s phrase “primal-domain expression.”

References

- [1] H.-C. Cheng and P.-C. Liu, *Error Exponents for Quantum Packing Problems via An Operator Layer Cake Theorem*, arXiv:2507.06232v4 (2026).
- [2] M. Nussbaum and A. Szkola, *The Chernoff lower bound for symmetric quantum hypothesis testing*, arXiv:quant-ph/0607216 (2006).
- [3] G. Androulakis and T. C. John, *Quantum f -divergences via Nussbaum–Szkola distributions and applications to f -divergence inequalities*, arXiv:2308.02929 (2023).
- [4] M. Sion, *On general minimax theorems*, Pacific J. Math. 8 (1958), 171–176.